NED University of Engineering and Technology

Computer Science and Information Technology

COMPLEX COMPUTING PROBLEM (CCP)
PROJECT SUBMISSION

Programming for Artificial Intelligence (CT-264)

Fairness-Aware Credit Default Prediction



Student Details

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1. Project Objective

To build and evaluate reproducible binary classifiers for predicting credit default, compare conventional predictive metrics before and after development-stage model/threshold tuning, and assess whether predictions and errors differ across demographic groups. The project additionally evaluates fairness mitigation approaches while preserving a held-out test set for final reporting.

2. Project Description

The study uses two tabular credit-risk datasets of different sizes. The target is binary: 1 denotes default and 0 denotes no default. Predictions of default are treated as a proxy for adverse credit screening, not as actual observed loan denials. The workflow separates data preparation, development-only cross-validation, hyperparameter/threshold selection, demographic fairness analysis, mitigation experiments, and an original-split final test evaluation. No fairness mitigation was applied to the selected final test models.

3. Executive Summary

Dataset	Selected model	Threshold	Test n	Accuracy	Precision	Recall	F1	ROC-AUC
German	Logistic Regression	0.40	151	0.6755	0.4730	0.7778	0.5882	0.8057
Taiwan	Random Forest	0.55	4,496	0.7903	0.5248	0.5538	0.5389	0.7802

The German model emphasizes recall of defaults, at the cost of more false positives. The Taiwan model achieves higher accuracy but lower default recall. Both models show age-related differences in predicted-default rates and false-positive rates. Cross-dataset comparisons are descriptive because prevalence, sample size and feature distributions differ.

4. Workflow and Research Design



Figure. End-to-end research workflow; tuning and selection occur before final test reporting.

5. Datasets and Preparation

Property	German Credit	Taiwan Credit Card Default
Raw rows	1,000	30,000
Clean rows	1,000	29,965
Original train / validation / test	700 / 149 / 151	20,975 / 4,494 / 4,496
Encoded predictor columns	54	23
Test defaults	45 / 151 (29.8%)	995 / 4,496 (22.1%)

Sensitive audit fields	Age, sex, foreign-worker	Age, sex

Numeric missing values were imputed using training-derived medians; categorical missing values used training-derived modes. Numeric scaling and one-hot category mappings were also fitted on the original training partition only. The credit amount was log-transformed in German preprocessing. Sensitive attributes were held in separate s_*.csv files and excluded from X features. Row identifiers were excluded from model predictors and used to verify X/y/s alignment.

Cleaning included dataset-specific data quality checks and duplicate handling. The Taiwan cleaned dataset contains 29,965 rows. The original 70/15/15 stratified split was preserved through the saved row IDs; reported sizes reflect integer rounding and the actual original split implementation.

6. Algorithms and Model Rationale

Method	Purpose and reason
Logistic Regression	Interpretable linear probability classifier; regularization C and balanced class weights; strong benchmark for structured credit data.
Random Forest	Nonlinear ensemble of decision trees; captures interactions and nonlinear effects; leaf-size control and class weighting reduce overfitting/imbalance effects.
Stratified 5-fold CV / OOF	Provides development-only out-of-fold predictions for fairer model/threshold comparisons and robustness checks.
Hyperparameter search	Compares predeclared candidate configurations on development folds; avoids tuning on final test data.
Decision-threshold tuning	Separates risk ranking from classification action; selected thresholds maximize development objectives subject to project choices.
Bootstrap 95% intervals	Quantifies uncertainty around development fairness differences, particularly important for small groups.
Fairness reweighing	Development-stage intervention to alter training emphasis across demographic/outcome combinations.
Group-specific thresholds	Development-stage post-processing experiment to reduce selected group outcome gaps; not deployed in final test.

A neural network and a from-scratch NumPy logistic-regression derivation appeared in the early project plan. The available completed-model evidence confirms Logistic Regression and Random Forest; no completed neural-network performance is substantiated here, so none is claimed.

7. Mathematical and Evaluation Framework

Logistic regression: $p(\text{default} \mid x) = 1 / (1 + \exp(-(w \cdot x + b)))$. The binary cross-entropy objective penalizes incorrect confident probabilities, with L2 regularization controlled by C. Random Forest averages predictions from multiple trees trained on bootstrap samples with randomized feature selection.

Accuracy = (TP+TN)/N; precision = TP/(TP+FP); recall = TP/(TP+FN); F1 = 2·precision·recall/(precision+recall). ROC-AUC measures ranking across thresholds; average precision summarizes the precision–recall curve and is informative with class imbalance. FPR = FP/(FP+TN); FNR = FN/(FN+TP). Predicted-default rate = (TP+FP)/group size.

7.1 Mathematical Foundations and Their Applications in the Project

The following formula images summarize the mathematics supporting the implemented pipeline. They explain underlying methods; the optimization and probability estimation were performed by scikit-learn, not manually calculated.

Linear Algebra

Customer features were represented as numerical vectors (54 encoded German features; 23 Taiwan features). The dot product combines features and learned coefficients into the logistic regression score. Feature scaling and one-hot encoding made those inputs suitable for the model.

$$z = \mathbf{w}^T\mathbf{x} + b$$

Linear algebra: weighted feature combination

Calculus and Optimization

Binary cross-entropy is the learning objective for logistic regression. A numerical optimization solver uses derivatives/gradient information to fit the coefficients. L2 regularization discourages overly large weights; the selected German model used C = 0.1. We did not implement manual gradient descent.

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_i + (1-y_i)\log(1-p_i)]$$

Calculus: logistic loss minimized by a numerical solver

Statistics

Stratified splitting maintained the default/non-default class proportions. Five-fold development cross-validation summarized predictive performance and its variability. Bootstrap 95% confidence intervals were used during the development fairness analysis to assess uncertainty. Train-fitted medians, modes, means and standard deviations supported preprocessing without fitting on test data.

$$\bar{s} = \frac{1}{K}\sum_{k=1}^{K} s_k \quad (K=5)$$

Statistics: average score across five CV folds

Probability

Logistic regression converts its linear score into an estimated default probability with the sigmoid function; random forest estimates class probabilities from tree predictions. Probabilities were converted to binary predicted-default labels using development-selected thresholds: German t = 0.40 and Taiwan t = 0.55. Thresholds were not changed after viewing test results.

$$p = \frac{1}{1+e^{-z}}, \quad \hat{y} = \mathbb{1}[p \geq t]$$

Probability: estimated default risk and locked threshold

Fairness and Statistical Group Comparisons

Group-level predicted-default rates, false-positive rates and false-negative rates were compared for age and sex; German foreign-worker status was also audited. In the corrected test results, the under-25 minus 25+ predicted-default-rate gap was approximately +9.99 percentage points (German) and +9.85 points (Taiwan). These are differences in model predictions, not observed loan denials or proof of discrimination.

$$\Delta_{PDR} = P(\hat{Y} = 1 \mid A = u) - P(\hat{Y} = 1 \mid A = o)$$

Fairness statistics: difference in predicted-default rates

How the mathematics supported evaluation

Confusion-matrix counts (TP, FP, TN, FN) were used to calculate accuracy, precision, recall, F1, FPR and FNR. ROC-AUC and average precision summarized ranking performance from probabilities. The final original-split German results were accuracy 0.6755, precision 0.4730, recall 0.7778 and F1 0.5882; Taiwan results were accuracy 0.7903, precision 0.5248, recall 0.5538 and F1 0.5389. Different class distributions mean accuracy alone should not determine the preferred model.

Methodological note

The first provisional test evaluation inadvertently reconstructed validation/test membership. The corrected evaluation used the saved original split IDs (151 German and 4,496 Taiwan test cases), retaining locked model choices and thresholds. Because provisional test results had already been viewed, the final evaluation should not be described as literally never inspected; no additional tuning was conducted in response.

8. Before Tuning: Initial Benchmarks

An early German development-stage 5-fold CV benchmark (849 development rows) reported the following mean ± fold standard deviation. These are preliminary, untuned reference results, not final held-out test estimates:

Metric	Logistic Regression baseline	Random Forest baseline
Accuracy	0.689 ± 0.033	0.722 ± 0.032
Precision	Not preserved in available benchmark	Not preserved in available benchmark
Recall	0.694 ± 0.077	0.643 ± 0.043
F1	0.572 ± 0.041	0.582 ± 0.037
ROC-AUC	0.772 ± 0.022	0.784 ± 0.032

A separate earlier German logistic-regression OOF run used all 1,000 rows at threshold 0.50 and reported accuracy 0.704, precision 0.505, recall 0.700, F1 0.587 and ROC-AUC 0.776. Because this run used a different data scope, it must not be treated as a directly comparable development-only pre-tuning baseline. Complete numerical baseline precision for both datasets and all candidate models was not available in the supplied records; the report deliberately does not invent those figures.

9. Hyperparameter Tuning and Locked Selection

Dataset / candidate	Best development CV configuration	CV ROC-AUC mean ± SD
German Logistic Regression	C=0.1; class_weight=balanced	0.7776 ± 0.0282
German Random Forest	300 trees; min_samples_leaf=5; balanced	0.7903 ± 0.0275
Taiwan Logistic Regression	C=1.0; class_weight=balanced	0.7483 ± 0.0103
Taiwan Random Forest	300 trees; min_samples_leaf=10; balanced	0.7800 ± 0.0057

Final selection was not based on ROC-AUC alone: development-stage F1, recall, interpretability, fairness evidence and threshold behavior informed the project choices. The locked final candidates were German Logistic Regression (C=0.1, balanced, threshold 0.40) and Taiwan Random Forest (300 trees, minimum leaf 10, balanced, threshold 0.55).

Selected model	Development OOF accuracy	Precision	Recall	F1	ROC-AUC	Average precision
German tuned LR + threshold	0.6572	0.4610	0.8353	0.5941	0.7742	0.5877
Taiwan tuned RF + threshold	0.7949	0.5346	0.5634	0.5486	0.7797	0.5528

The early untuned and final tuned figures above are from different reporting stages and should not be interpreted as controlled, same-fold before/after differences. In particular, German baseline precision and Taiwan untuned precision/accuracy are not fully preserved. The reliable controlled comparison is the saved development model-comparison report; its complete baseline rows should be inserted if later supplied.

10. Final Original-Split Test Evaluation

The selected models were refitted on the concatenated original training and validation feature matrices (German 849; Taiwan 25,469 rows). The preprocessing transformations themselves remained fitted on the original training split, rather than being refitted. The original saved X/y/s row IDs were validated for alignment and non-overlap. Test membership and thresholds were not reconstructed or changed during the corrected evaluation.

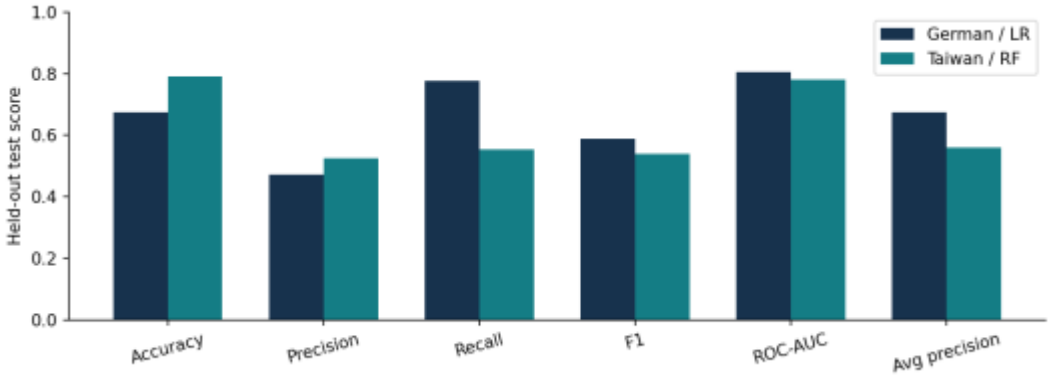


Figure. All six held-out test performance metrics; German and Taiwan are distinct datasets.

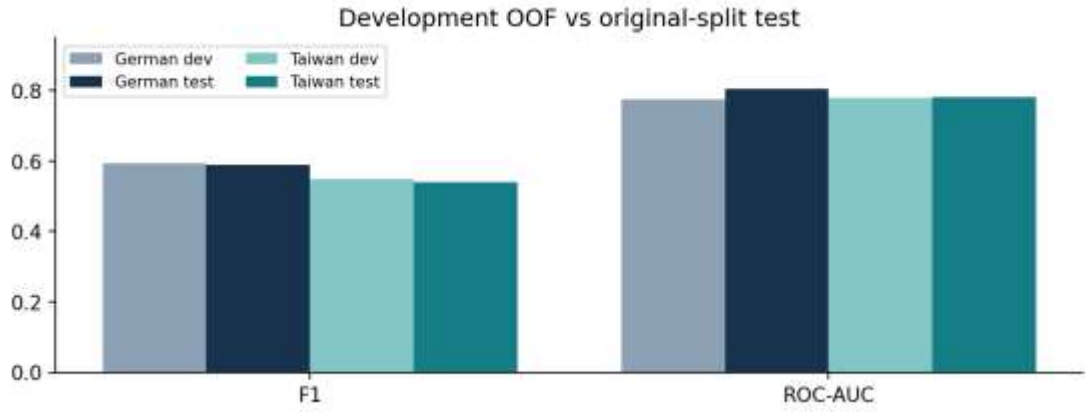


Figure. Locked development OOF metrics compared with the corrected original-split test metrics.

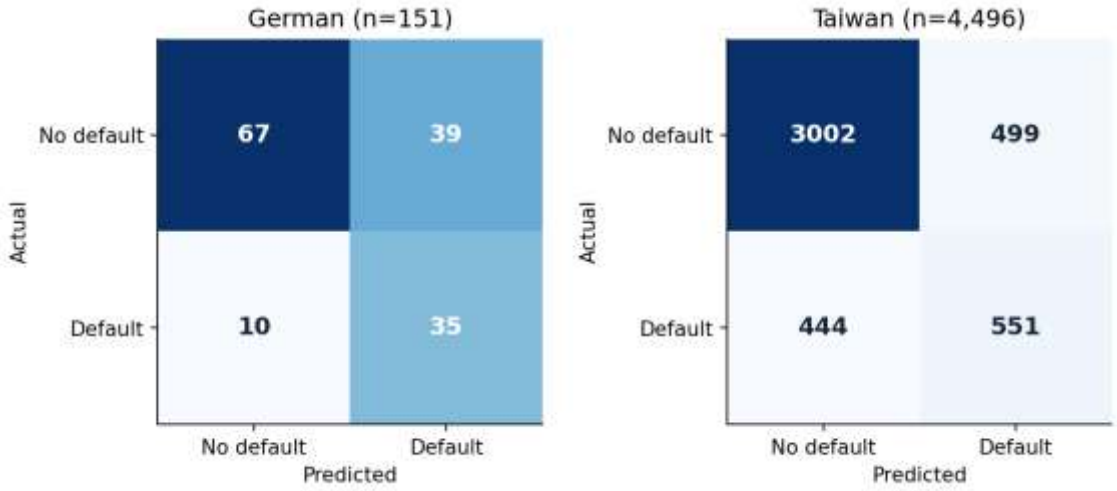


Figure. Test confusion matrices; rows are actual classes and columns are predicted classes.

Test outcome	German	Taiwan
True negatives	67	3,002
False positives	39	499
False negatives	10	444
True positives	35	551
Precision	0.4730	0.5248
Recall	0.7778	0.5538
F1	0.5882	0.5389
ROC-AUC	0.8057	0.7802
Average precision	0.6749	0.5606

German correctly identified 35 of 45 actual defaults but also flagged 39 non-defaulters. Taiwan correctly identified 551 of 995 actual defaults and missed 444. Both false positives and false negatives have real-world relevance; the desirable tradeoff depends on policy and the cost of each type of error.

11. Final Test Fairness Audit

Sensitive features were not used as predictors. Group fairness was audited after the final predictions were frozen. Differences in predicted-default rates are not observed loan rejection rates, and unequal outcomes alone do not prove causation or unlawful discrimination.

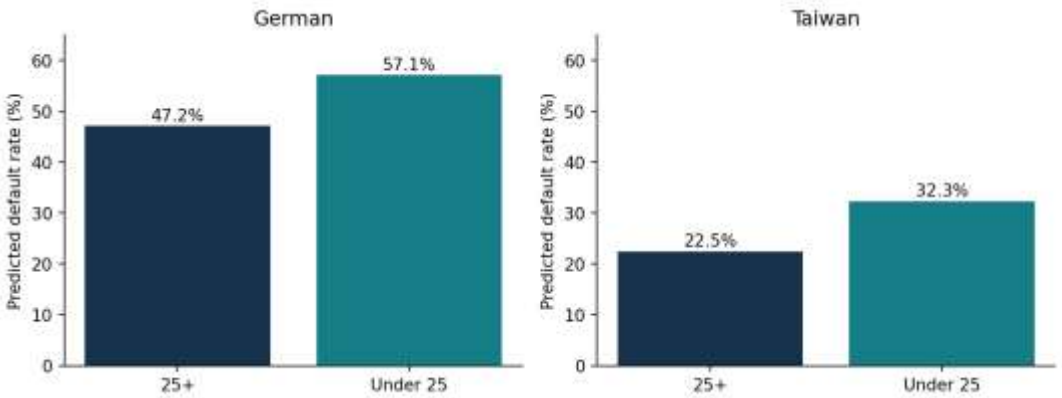


Figure. Predicted-default rates by age group on the corrected held-out test.

Dataset / age group	n	Actual default	Predicted default	FPR	FNR
German 25+	123	26.02%	47.15%	35.16%	18.75%
German under 25	28	46.43%	57.14%	46.67%	30.77%
Taiwan 25+	4,097	21.82%	22.48%	13.61%	45.75%
Taiwan under 25	399	25.31%	32.33%	21.14%	34.65%

Age predicted-default gaps (under 25 minus 25+) were +9.99 percentage points in German and +9.85 points in Taiwan. Corresponding FPR gaps were +11.51 and +7.53 points. Group base rates also differ; these figures require contextual interpretation.

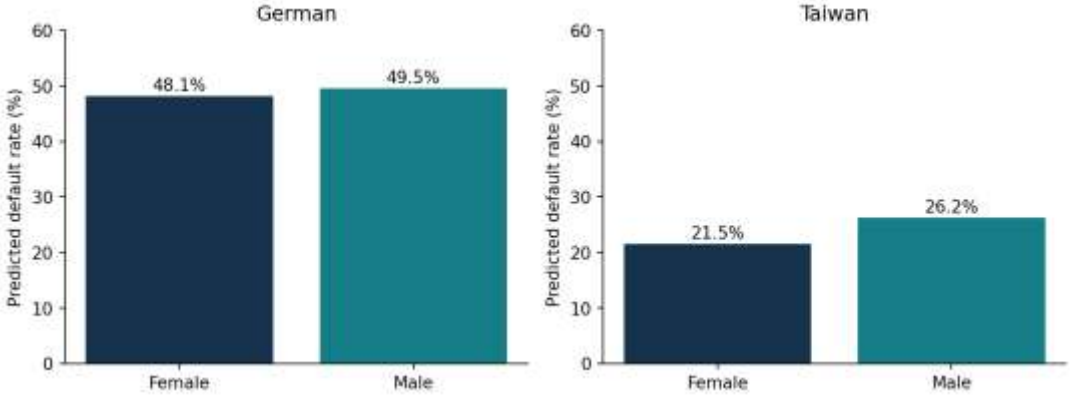


Figure. Predicted-default rates by sex on the corrected held-out test.

Dataset / sex	n	Predicted default	FPR	FNR
German female	54	48.15%	31.43%	21.05%
German male	97	49.48%	39.44%	23.08%
Taiwan female	2,721	21.50%	12.74%	44.22%
Taiwan male	1,775	26.20%	16.72%	45.12%

Sex predicted-default gaps (male minus female) were +1.34 points in German and +4.70 points in Taiwan. The German foreign-worker=no test subgroup comprised only five records (predicted-default rate 20%); the yes subgroup comprised 146 (50%). The 30-point raw gap is not stable evidence because n=5 is far too small for reliable subgroup inference.

12. Development Fairness Mitigation Experiments

Mitigation was evaluated on development data, not on the final test model. Reweighing and group-specific threshold adjustments were explored; the threshold results below show the tradeoff between reducing a selected disparity and preserving predictive performance. They are simulations, not deployment recommendations.

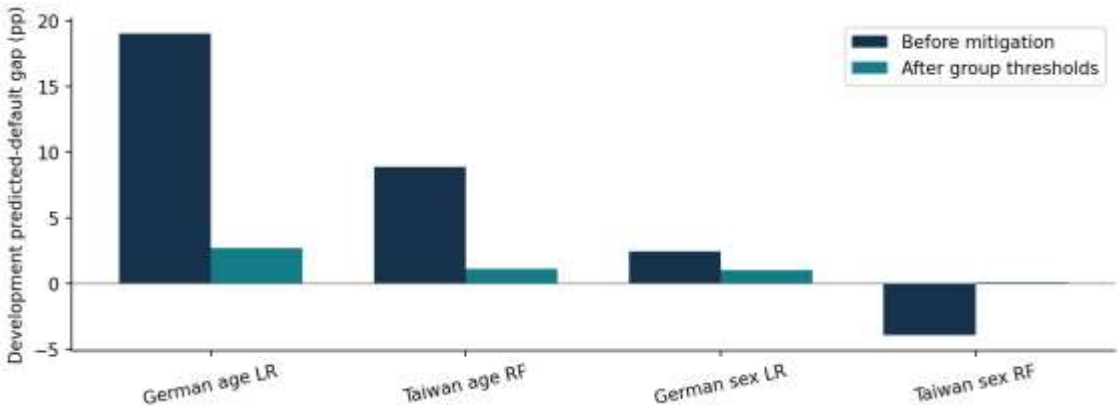


Figure. Development-only predicted-default rate gaps before and after group-specific threshold adjustment; signed values preserve direction.

Development scenario	F1 before → after	Predicted-default gap before → after
German age / LR	0.5946 → 0.5831	+0.1904 → +0.0271
German age / RF	0.5919 → 0.5739	+0.2094 → +0.0284
Taiwan age / LR	0.5228 → 0.5081	+0.1313 → −0.0055
Taiwan age / RF	0.5486 → 0.5349	+0.0890 → +0.0113
German sex / LR	0.5946 → 0.5805	+0.0243 → +0.00996
German sex / RF	0.5919 → 0.5862	+0.0112 → +0.00518
Taiwan sex / LR	0.5228 → 0.5132	−0.0375 → +0.0092
Taiwan sex / RF	0.5486 → 0.5423	−0.0393 → +0.00079

For the selected German LR model, the development age gap fell from 19.04 to 2.71 percentage points, while F1 fell from 0.5946 to 0.5831. For the selected Taiwan RF model, the age gap fell from 8.90 to 1.13 points, while F1 fell from 0.5486 to 0.5349. Group-specific thresholds raise operational, legal, and ethical questions and were not used in final evaluation.

13. Challenges, Corrections and Resolution

Challenge / limitation	How addressed or documented
Class imbalance	Stratified splits, balanced class weights, precision/recall/F1/AP alongside accuracy.
Data leakage risk	Training-only preprocessing fit; X/y/s row ID checks; test excluded from development tuning.
Reconstructed split mismatch	Detected validation/test mismatches; switched to original saved CSV row IDs; corrected test sizes to 151 and 4,496.
Circular import / script mix-up	Restored module separation and ran the explicitly corrected evaluation script.
Initial provisional test results already viewed	Preserved outputs; corrected protocol without retuning; documented that the final corrected test was not a never-before-viewed first test exposure.
Small demographic subgroups	Reported counts and warned against strong inference, especially German foreign_worker=no (n=5).
Performance–fairness tension	Reported development mitigation F1 and disparity changes side by side.

Incomplete archived baseline precision	Marked unavailable values explicitly; did not manufacture missing results.
Proxy features and residual bias	Excluded direct sensitive columns, audited outcomes, and noted that proxy associations can remain.

14. Limitations and Responsible Interpretation

- Observational datasets do not establish causal discrimination; group gaps may reflect base-rate differences, proxies, measurement and sampling effects.
- The German test set (151 records) and small subgroups produce imprecise fairness estimates; formal uncertainty intervals for the corrected test fairness figures are not provided here.
- The Taiwan and German datasets differ in size, populations and variables; higher cross-dataset accuracy does not establish a universally superior algorithm.
- Group-specific threshold mitigation was evaluated only as a development simulation; no final held-out mitigation claims are made.
- Training-fitted preprocessing was reused when fitting final models on train+validation, rather than refitting the transformation on all development rows.
- An earlier provisional test run exposed test results before original-split correction. No model or threshold was retuned afterward, but the process is not a pristine single-look test protocol.
- The available evidence supports two implemented final classifiers, not a completed neural-network comparison or a deployed credit decision system.

15. Conclusion

This project demonstrates a reproducible, fairness-aware evaluation workflow for credit default prediction across two datasets. The locked German Logistic Regression classifier achieved 67.55% accuracy, 47.30% precision, 77.78% recall, F1 0.5882 and ROC-AUC 0.8057 on 151 original held-out rows. The locked Taiwan Random Forest achieved 79.03% accuracy, 52.48% precision, 55.38% recall, F1 0.5389 and ROC-AUC 0.7802 on 4,496 original held-out rows. Both datasets exhibited higher predicted-default and false-positive rates for under-25 borrowers. Development-only mitigation reduced selected group disparities, typically with modest F1 costs. The central finding is that predictive utility, subgroup error behavior and procedural evaluation integrity must be assessed together; no single metric captures the full credit-risk decision tradeoff.

16. Reproducibility and Output Inventory

Key project modules: src/preprocess.py, src/train.py, src/cv.py, src/fairness_compare.py, src/fairness_bootstrap.py, src/fairness_mitigation.py, src/threshold_fairness_mitigation.py, src/compare_models.py, src/final_model_selection.py and src/final_test_evaluation_fixed.py. Original processed split files: data/processed/{german,taiwan}/{X,y,s}_{train,val,test}.csv. Locked model selection: reports/final_model_selection/selected_models.json. Corrected final outputs: reports/final_test_evaluation_original_splits/{german,taiwan}/, containing metrics.json, predictions.csv, model.joblib, confusion_matrix.png, roc_curve.png, precision_recall_curve.png and group fairness CSVs.

Figures in this report are redrawn from the supplied numerical outputs. They are not claimed to be the exact original ROC/precision–recall curve images; those original files remain in the local project report folders and can be inserted in a future revision.

Appendix A. Suggested Final Figures from Local Project

For a fully archival submission, replace or supplement the report's summary graphics with the actual exported confusion_matrix.png, roc_curve.png and precision_recall_curve.png for each dataset, plus selected development-stage model comparison and fairness bootstrap plots. These original plot files were not supplied to this document-generation session.

Appendix B. Data and Provenance Notes

Numerical results and pipeline details are drawn from the user-provided terminal outputs, project source excerpts, and the conversation’s confirmed development metrics. Unavailable original baseline rows and unsupplied original plot images are clearly distinguished from verified results. The report intentionally avoids fabricating test fairness confidence intervals or neural-network results.